Software Engineering & Project

Snapshot Week 10 - BLOCK UG-16

Maptek, Block Model Compression Algorithm

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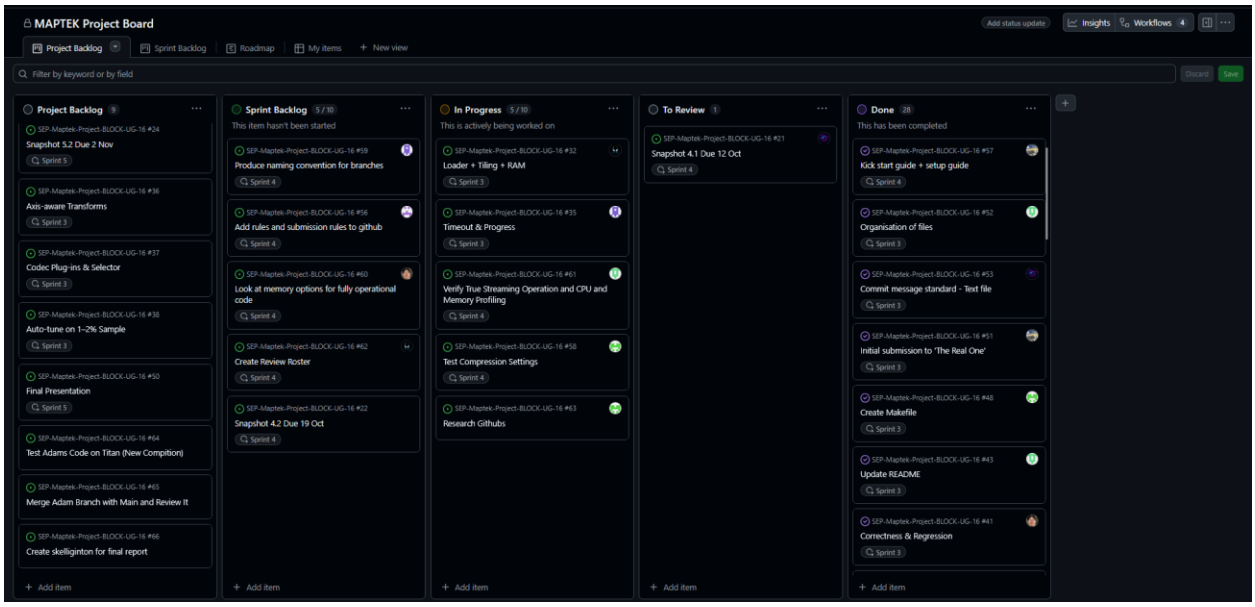
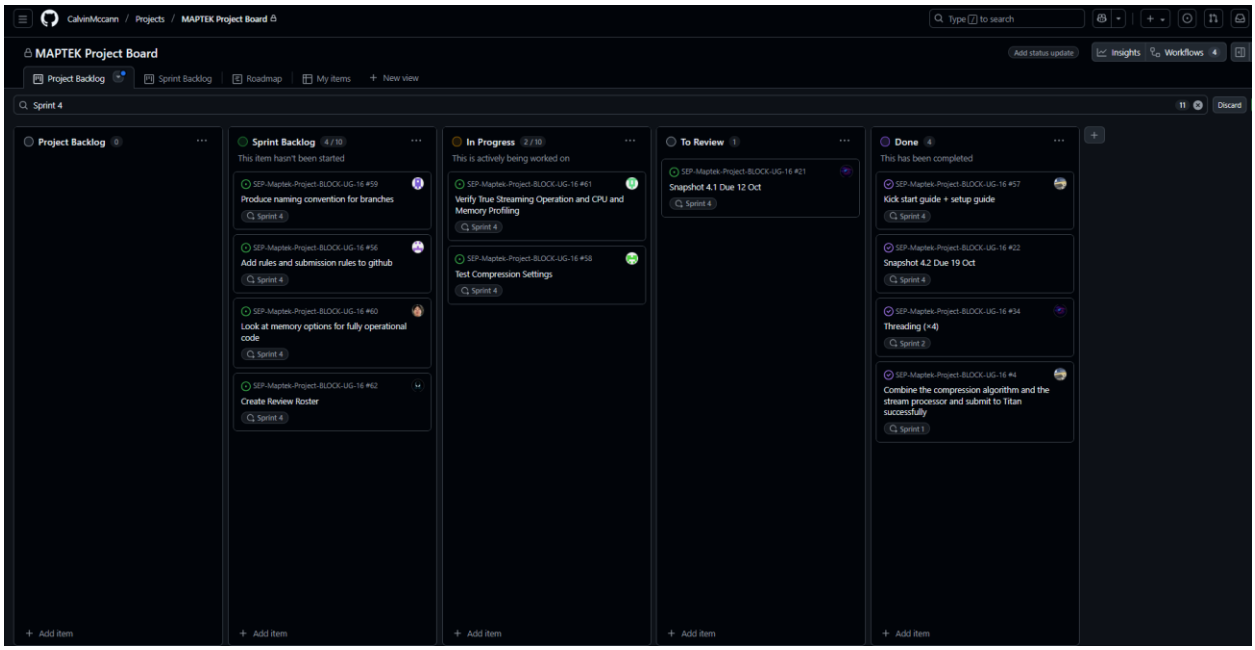
Product Backlog and Task Board

Product Backlog	
Task	**Description**
Snapshot 4.1	Create snapshot 4.2
Snapshot 4.2	Create snapshot 4.2
Snapshot 5.1	Create snapshot 5.1
Snapshot 5.2	Create snapshot 5.2
Axis-aware Transforms	delta/RLE per axis Palette/dictionary per tile
Correctness and Regression	Do: run current code on intro and fast competitions; confirm outputs and acceptance still pass. Done when: both older comps pass their acceptance checks.
Docs & Results Sheet	Shared results sheet (commit, params, ratio, MB/s, RAM, runtime)
Final Presentation	Initial planning for final presentation
Test Adams Code on Titan (New competition)	Test Adam code for competition 4
Merge Adam Branch with Main and Review It	Review Adams branch and merge it with main
Create Skeleton for final report	Create skeleton for final report.



Sprint Backlog and User Stories

Sprint Backlog	
Task	**Description**
Test Compression Settings	Reduce compression settings and create either a verbal report or written report
Verify True Streaming Operation and CPU Profiling	Verify compressor reads block by block and find where time and resources are being spent throughout code
Check Memory Options	Find variables in main code and change to reduced option
Create Review Roster	Create review roster for branches being merged to main
Produce naming convention for branches	Create naming convention file for branches
Add rules and submission rules to GitHub	Add rules and submission rules to GitHub



User Stories (Sprint 4)

For a user of the Block Model Compression Algorithm, this sprint focused on stability, maintainability, and preparation for the final competitions — *The Streaming One* and *The Worldly One*. Following the successful implementation of true streaming functionality in the previous sprint, the team aimed to refine performance, optimise memory usage, and improve collaboration and documentation workflows.

The sprint involved testing and tuning compression settings to achieve a balanced trade-off between compression ratio and runtime performance. These adjustments were documented in a short internal report to guide future parameter selections. To ensure the algorithm's correctness and reproducibility, CPU profiling and streaming verification tasks were performed to confirm that data is read and processed block-by-block, without preloading entire datasets. The profiling results provided insight into where time and memory resources are being consumed, informing future optimisation strategies.

The team also worked on memory management improvements, identifying key variables in the main code that could be adjusted to

reduce the algorithm's memory footprint during streaming. This ensures smoother operation on large datasets and aligns with the requirements of *The Streaming One* competition.

Beyond technical improvements, the sprint also introduced several workflow and collaboration enhancements. A review roster was created to define responsibilities for reviewing and approving code before merging to the main branch. A branch naming convention file was added to the repository to enforce consistent and organised version control practices. Additionally, GitHub submission rules were documented to clearly define contribution standards and ensure quality and traceability of submissions.

Together, these updates strengthen both the technical reliability and collaborative efficiency of the team, setting the foundation for final testing, presentation preparation, and competition submissions in the upcoming sprint.

Definition of Done

- Compression settings tested and documented for speed vs ratio trade-offs.
- Verified that the compressor operates as a true streaming system (block-by-block).
- CPU profiling and memory usage reports completed and reviewed.
- Memory optimisation parameters identified and applied where beneficial.
- Review roster created and implemented for main branch merges.
- Branch naming convention file added to repository.
- GitHub submission rules finalised and published for all contributors.

Summary of Changes

During Sprint 5, the team focused on consolidating and optimising the algorithm:
Reduced and tested compression settings to improve efficiency.
Verified true streaming operation and conducted detailed CPU/memory profiling.
Improved memory management through variable and parameter adjustments.
Established review rosters and branch naming conventions for consistent workflow.
Added GitHub submission and contribution rules to standardise team practices.

By the end of this sprint, the algorithm achieved greater stability, resource efficiency, and process maturity, ensuring readiness for the final phase — maximising compression performance and preparing for presentation and submission in Week 11.